GEOL540	ISOTOPE GEOCHEMISTRY AND GEOCHRONOLOGY	3.0
5. At least One elective approved by your advisor.		
Total Credits Required for MS:		**30.0**
Total Credits Required for PhD:		**72.0**

The student's thesis committee may specify additional course requirements and makes final decisions regarding transfer credits.

Students who enter the PhD program with a thesis-based master's degree may transfer up to 36 credits in recognition of the coursework and research completed for that degree. At the discretion of the student's thesis committee, up to 24 credits of previous graduate-level coursework (at Mines or elsewhere) can be applied toward the course requirement of the Doctor of Philosophy (Geochemistry degree track) program.

A total of 18 course credits are required with at least 9 credits being completed at Mines.

Comprehensive Examination

Doctor of Philosophy (Geochemistry) students in both degree tracks must take a comprehensive examination. It is expected that this exam will be completed within three years of matriculation or after the bulk of coursework is finished, whichever occurs earlier. This examination will be administered by the student's thesis committee and will consist of an oral and a written examination, administered in a format to be determined by the thesis committee. Two negative votes in the thesis committee constitute failure of the examination.

In case of failure of the comprehensive examination, a re-examination may be given upon the recommendation of the thesis committee and approval of the Dean of Graduate Studies. Only one re-examination may be given.

Mines' Combined Undergraduate/Graduate Degree Program

Students enrolled in Mines' combined undergraduate/graduate program may double count up to 6 credits of graduate coursework to fulfill requirements of both their undergraduate and graduate degree programs. These courses must have been passed with B- or better, not be substitutes for required coursework, and meet all other university, department, and program requirements for graduate credit.

Students are advised to consult with their undergraduate and graduate advisors for appropriate courses to double count upon admission to the combined program.

Graduate Certificate of Analytical Geochemistry

This program offers an opportunity for working professionals to complete graduate level coursework leading to a Graduate Certificate in a short time. The program focuses on providing instruction in a.) the fundamentals of geochemical analysis, which give students the flexibility to respond to changing opportunities in earth and environmental science, and b.) the mechanics of common techniques (sample collection and preparation, XRF, SEM, and ICP), which prepare students for immediate entry into the jobs market. The Certificate program is comprised of 9.0 credits of coursework. Please note that courses cannot be double counted to fulfill the requirements of other Certificates.

Students are required to take at least 6.0 credits out of the following core courses:

Core Courses:		
CHGC504	METHODS IN GEOCHEMISTRY	3.0
GEGN532	GEOLOGICAL DATA ANALYSIS	3.0
GEOL523	REFLECTED LIGHT AND ELECTRON MICROSCOPY *	2.0
GEOL540	ISOTOPE GEOCHEMISTRY AND GEOCHRONOLOGY	3.0

Students can choose up to 3.0 credits out of the following elective courses:

Electives:		
CHGC503	INTRODUCTION TO GEOCHEMISTRY	3.0
CEEN560	MOLECULAR MICROBIAL ECOLOGY AND THE ENVIRONMENT	3.0
CEEN562	ENVIRONMENTAL GEOMICROBIOLOGY	3.0
CHGN583	PRINCIPLES AND APPLICATIONS OF SURFACE ANALYSIS TECHNIQUES	3.0
GEGN586	NUMERICAL MODELING OF GEOCHEMICAL SYSTEMS	3.0
GEGN587	HYDROCHEMICAL AND TRANSPORT PROCESSES	3.0
GEGX571	GEOCHEMICAL EXPLORATION	3.0
GEOL513	HYDROTHERMAL GEOCHEMISTRY	3.0
GEOL628	ADVANCED IGNEOUS PETROLOGY	3.0
MTGN605	ADVANCED TRANSMISSION ELECTRON MICROSCOPY	2.0
PHGN504	RADIATION DETECTION AND MEASUREMENT	3.0

* Students can add one additional credit of independent study (GEOL599) for XRF methods which is taken concurrently with GEOL523.

Professional Masters' Degree in Analytical Geochemistry

This program is designed for both full time students and working professionals who want to increase their knowledge and skills, while gaining a thorough up-date of advances across the spectrum of geochemical analysis.

A minimum of 30 credits are required for the PM degree. Students working towards a PM in Analytical Geochemistry are required to take at least 9.0 credits from courses in the Certificate in Analytical Geochemistry core and 3.0 credits from the electives. Up to 6 credits of the additional electives can include an independent study project (CHGC 598) at Mines, federal agencies, or industry, and is highly encouraged.

The Certificate in Analytical Geochemistry can be combined with the Certificate in Exploration Methods to be used towards the Professional Masters in Analytical Geochemistry (Non-thesis), which will allow part-time students to stack their education. When stacking the Certificates, the additional 6 credits required to complete the PM can include an independent study project (CHGC 599) at Mines, federal agencies, or industry, and is highly encouraged. Independent study projects that connect students with local entities will be a priority and will establish a network of future employers for the students and collaborations for CSM.